What Is Naturalness?

Williams's paper is organized around a single question—*what is the naturalness principle, and is it well-motivated?*—against the backdrop of the 2012 discovery of a 125 GeV Higgs with no accompanying new physics, which put the principle under pressure.

He defends treating naturalness as a *prohibition on sensitive correlations between widely separated physical scales*. He argues (i) this single idea unifies the discordant formulations found in the physics literature, (ii) it is well-grounded both theoretically (Decoupling Theorem, RG) and empirically (past heuristic success), and (iii) its apparent failure in the Standard Model is a genuine *empirical* challenge — with further consequences, in particular undermining the picture of nature as a hierarchy of quasi-autonomous domains.

Williams surveys five competing characterizations of naturalness, each of which he argues to be flawed (but partially reflect his preferred account). The point of the survey is diagnostic: physicists agree there *is* a problem but describe it in ways that variously mislead, obscure its physical content, or invite illegitimate dismissal.

Five Faces of Naturalness

The setup is the Hierarchy Problem. In a Yukawa theory with a heavy fermion Ψ (mass M) and a light scalar ϕ, integrating out the fermion to obtain a low-energy EFT for ϕ generates a correction to the scalar mass quadratic in the cutoff (Λ):

$$(m^*)^2 = m^2 + \frac{g}{16\pi^2}\left[\Lambda^2 + M^2 + m^2 \ln \tfrac{\Lambda}{M}\right] + \cdots$$

The mass of the "light" scalar is driven up to the cutoff scale — contradicting the premise that it belongs to the low-energy theory. (Contrast: fermion masses receive only multiplicative corrections $\propto M$, so a light fermion stays light.) The five formulations are competing diagnoses of what exactly has gone wrong.

1. **Quadratic divergences.** *The claim:* the problem is that scalar-mass corrections diverge as $\Lambda \to \infty$ (typical textbook framing). *Williams's objection:* two defects. (a) From an EFT standpoint there is no reason to take $\Lambda \to \infty$; better to speak of quadratic *sensitivity* to the finite scale Λ. (b) More seriously, the quadratic divergence is *regulator-dependent*: redo the calculation in dimensional regularization and no Λ^2 term appears at all—yet a correction $\Delta m^2 \sim M^2$ (quadratic in the *heavy fermion mass*) survives. Since changing the regulator removes the divergence but not the problem, quadratic divergences cannot have been the real issue.

2. **Symmetry principle ('t Hooft).** *The claim:* a small parameter is natural iff setting it to zero increases the symmetry of the theory. (The light fermion is "protected" by the chiral symmetry restored at $M \to 0$; the scalar mass enjoys no such protection.) *Williams's objection:* this gives only a *sufficient* technical condition, not the physical content of naturalness—and it is not *necessary*. The paradigm case of a naturally small parameter, the QCD scale Λ_{QCD}, owes its smallness to *dimensional transmutation* (a feature of QCD dynamics), not to any symmetry. Defining naturalness via symmetry would force one either to call a paradigmatically natural scenario "unnatural" or to add on a separate account for dynamically-secured cases.

3. **Fine-tuning.** *The claim:* unnatural theories require improbably precise cancellations—one must tune the bare mass to ~34 decimal places to keep the Higgs light. *Williams's objections:* (a) It *lumps naturalness together with unrelated fine-tuning problems* (e.g. the flatness, horizon, and low-entropy problems in cosmology), which involve no correlation between separated scales; by contrast, the "unnaturalness" of a parameter is measured precisely by the *ratio of scales* over which its low- and

high-energy values must be correlated. (b) The renormalization rejoinder—"just absorb the corrections into a renormalization of the bare mass"—fails, because integrating out is *recursive*: the mass can be renormalized at only one scale, yet receives a fresh large correction at every mass threshold the RG flow crosses, so the sensitivity is ineliminable. (c) Framing naturalness as mere fine-tuning makes for a *weak* argument, since a popular pragmatic reading of EFT (treat low-energy parameters as empirical inputs, arrange high-energy parameters as needed) supports rejecting this as a problem.

4. **Aesthetic / sociological criterion.** *The claim:* naturalness is a theorists' taste or prejudice, not a "normal scientific argument" (Grinbaum, Shifman, more recently Hossenfelder). *Williams's objections:* (a) The aesthetic reading is undercut by the fact that the most popular natural solution, low-energy SUSY, is widely conceded to be *inelegant* (124 new constants); naturalness and beauty come apart. (b) The sociological case trades on the arbitrariness and historical drift of *formal fine-tuning measures* (acceptable tuning has crept from $\sim$10 toward 1000 over the years). But these measures are merely poor formalizations: their stated *physical* motivation is exactly the prohibition of interscale correlation, and they notoriously misfire (assigning high fine-tuning to paradigmatically natural cases like Λ_{QCD}). "Throwing away the ladder" and fixating on the numbers is what makes naturalness look merely aesthetic.

5. **Prohibition on interscale sensitivity (the view Williams advocates).** Low-energy physics should not depend sensitively on a theory's description of physics at much higher energies. This (a) underwrites the legitimate core of each rival formulation (quadratic sensitivity is the special case where the relevant high scale is Λ itself; symmetry is one sufficient mechanism for securing insensitivity; the "informal" fine-tuning measure just *is* the scale ratio), while (b) discarding their defects. The RG makes the criterion precise: a *relevant* coupling at a low scale Λ_L is extraordinarily sensitive to its value at a high scale Λ_H (altering $\lambda_2(\Lambda_H)$ in the 20th decimal place shifts $\lambda_2(\Lambda_L)$ by a factor of 10^8), whereas *irrelevant* couplings are essentially independent of their high-energy initial values. Unnaturalness $=$ ineliminable sensitivity of a relevant coupling to far-separated scales.

Background: Effective Field Theories

- An EFT is a QFT with a UV cutoff Λ excluding field modes above Λ. Why treat *all* empirically applicable QFTs as EFTs? Gravity is not perturbatively renormalizable (no QFT is trustworthy to arbitrarily high energy), and non-asymptotically-free theories (QED, ϕ^4) hit Landau poles at finite energy. (Landau pole: coupling constants are not truly constant, they "ru with energy" and *diverge* at finite energy, signaling at very least the breakdown of perturbation theory.)

- The Wilsonian effective action contains infinitely many operators; dimensional analysis sorts them into *relevant* (dim < 4, coefficients $\propto$ positive powers of Λ, contributions grow toward low energy), *marginal* (dim $= 4$), and *irrelevant* (dim > 4, suppressed by Λ).

- **RG:** integrating out modes $\mu \leq |\mathbf{k}| \leq \Lambda$ yields a new EFT below μ with modified ("running") couplings. *All* sensitivity to the integrated-out high-energy physics is carried by the change in the couplings.

The Decoupling Theorem (Appelquist–Carazzone 1975)

This theorem is doing much of the load-bearing work in the paper—both in grounding the "central dogma" that scales decouple and in the later discussion of consequences.

- **Statement.** Start from a *perturbatively renormalizable* "full theory" containing fields of mass $\gg$ the rest. Then the correlation functions for physics at energies far below the heavy masses can be reproduced by an EFT containing *only the light fields.* The sole remnant of the heavy fields is a modification of the light-field couplings.

- **The fermion/gauge-boson case.** If the theory contains only fermions or gauge bosons, no relevant operators appear and the heavy fields' contribution is merely *logarithmic*—a small correction. Virtually every QFT in particle physics was of this form until the Higgs: *no elementary scalar had been included.* The scalar is exactly what introduces a relevant operator (its mass term) that receives large corrections from high-energy physics.

- **Why the theorem is surprising.** Loop corrections to correlation functions automatically include contributions from field modes at *all* momentum scales in the theory's domain. That all such high-energy contributions can be *confined to shifts in a finite set of couplings*—rather than leaking into the low-energy physics more directly—is a striking result.

- **The two crucial provisos** (these matter decisively in § on ontology): the proof requires *(i)* a perturbatively renormalizable starting point, and *(ii)* a *mass-dependent* renormalization scheme. In the mass-*independent* schemes that are usually most convenient (dimensional regularization + minimal subtraction), the theorem *cannot be proved* and decoupling is not manifest—it must be "put in by hand," a move justified only *empirically*, by the accuracy of the resulting predictions.

- **Relation to naturalness.** The Higgs does *not* strictly violate the Decoupling Theorem (its high-energy sensitivity shows up exactly where the theorem permits—as a modification of the mass coupling). It violates instead the *spirit* of decoupling: it leads to finely-correlated sensitivity to high energy scales rather than a benign logarithmic shift.

The Central Dogma, and Why Failures of Naturalness Matter

- **Central dogma of EFT:** widely separated scales should largely decouple. *Theoretical support:* the Decoupling Theorem and the successes of the RG (irrelevant couplings forget their high-energy initial values). *Empirical support:* (i) the spectacular precision of the Standard Model entails that unknown high-energy physics has little effect on probed low-energy observables; (ii) several historical cases where enforcing decoupling had heuristic value (cf. Wells).

- **Are unobservable correlations worth worrying about?** Williams: yes. (i) Even when unobservable, the sensitivity affects the RG flow of the renormalized, physical mass, which "jumps" at every mass threshold unless the initial renormalization is chosen to cancel every later correction. (ii) Methodologically, solving naturalness problems has been a good heuristic guide.

- **Success: the charm quark.** Gaillard & Lee (1974), computing the K_L^0–K_S^0 mass difference in a kaon-scale EFT, faced two options consistent with experiment: (i) strange and charm masses both large but finely cancelling, or (ii) the charm mass alone setting the scale, requiring $m_c \sim 1.5$ GeV. They rejected the "fortuitous cancelation" (an interscale-correlation, hence unnaturalness, argument) and predicted (ii). Charm was found in 1974 at ~ 1.3 GeV.

- **Failure: the cosmological constant.** A dimension-zero (relevant) operator requiring tuning even more delicate than the Higgs; no natural solution is in sight ("pragmatic despair"). A standing reminder that naturalness expectations *can* be mistaken.

Quasi-Autonomous Domains?

Consider proposal that EFTs reveal nature as "layered into quasi-autonomous domains, each layer having its own ontology and associated 'fundamental' laws" (Cao & Schweber 1993), where the stability and independence of each level is taken to *follow from* the Decoupling Theorem. (Williams leaves "quasi-autonomous" deliberately vague: any precisification strong enough to support the intended ontological hierarchy will do.)

- **The Cao–Schweber argument.** The Decoupling Theorem explains why a description at one level is stable and undisturbed by higher-energy happenings, thereby underwriting a hierarchy of independent ontological levels. (Their further claims— ontological pluralism, epistemic antifoundationalism, methodological antireductionism— Williams sets aside as overreach, following Huggett & Weingard 1995.)

- **The Hartmann (2001) and Bain (2013) critiques (of the *general* claim).** The Decoupling Theorem's two provisos can fail. *Hartmann:* the proof needs a renormalizable starting point, which an unending tower of (generically nonrenormalizable) EFTs will not provide—this is in tension with Cao–Schweber's own "antireductionism." *Bain:* the proof needs a mass-dependent scheme; in mass-independent schemes the theorem is unavailable and decoupling is inserted by hand. So the Decoupling Theorem cannot, *in general*, underwrite a quasi-autonomous-domains ontology.

- **But our actual world looks like a candidate.** The Standard Model *is* perturbatively renormalizable with widely separated mass scales, and so satisfies the theorem's conditions; alternatively, decoupling can be put in by hand and justified empirically. So prima facie one *can* read off a hierarchy of quasi-autonomous domains—and Bain (2013) and Castellani (2002) tentatively do.

- **Williams's distinctive move: naturalness, not the theorem, is what fails.** The Standard Model contains *two relevant parameters*—the Higgs mass and the cosmological constant—whose low-energy values are extraordinarily sensitive to the cutoff scale (the $10^{-20} \to 10^{8}$ RG sensitivity). Whatever "quasi-autonomy" means, it must *prohibit* this kind of extreme interscale dependence between supposedly independent levels. So even though the Decoupling Theorem holds for the Standard Model, its *unnaturalness* undermines the quasi-autonomy that the theorem appeared to license.

- **Why this is stronger than Hartmann/Bain.** Their critiques block only the *general* inference from the theorem; they leave open that our world supports the ontology. Williams blocks that residual route too—and crucially, failures of naturalness are *scheme-independent* (the relevant-parameter sensitivity does not vanish on switching to a mass-independent scheme). So Bain's own positive proposal— that continuum EFTs (mass-independent scheme, decoupling by hand) support quasi-autonomous domains—also fails: the sensitivity is there regardless.

- **Upshot (methodological).** The Decoupling Theorem, *even where it holds*, is too weak to underwrite an ontology of quasi-autonomous domains. An additional necessary condition is that the EFT be *natural*. Any attempt to ground such an ontology in features of the EFT framework is "doomed to fail unless the theory in question is natural."

References

Williams, P. (2015). Naturalness, the autonomy of scales, and the 125 gev higgs. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 51:82–96.